

Software for Mobile Devices (CS4039)

Date: May 18, 2026

Course Instructor(s)

Mr. Rana Waqas Ali

Final Exam (Subjective)

Total Time:

2 H 30 M

Total Marks:

50

Total Questions:

5

226-7983

Roll No

BSE-88

Section



Student Signature

Read **INSTRUCTIONS** carefully before starting exam

- **Objective Part** (distributed separately) shall be attempted in first 30 minutes only.
- **Objective Part** shall be attached with the answer book
- This is a closed book exam.
- **Attempt all questions in sequence on the answer sheet.**
- Irrelevant or extra code will lead you to get **negative marks**.
- **No need to write code in XML. Only JAVA and KOTLIN** are allowed for this exam.

[10 Marks]

[CLO-1] Question 2: Volley Library + JSON Parsing + RecyclerView.

You are developing a **News Android Application**. The application fetches news data from an online JSON API using Volley library and displays it in RecyclerView.

Each news object contains:

- title
- description

Requirements:

- Write Java code to create a Volley `RequestQueue`.
- Write Java code using `JsonArrayRequest` to fetch JSON data from an API URL.
- Inside `onResponse()`, parse:

- title
- description

from JSON response and store them into an `ArrayList`.

- After parsing data, notify RecyclerView Adapter to update the UI. Write required code.

[CLO-3] Question 3: SQLiteDatabase + SharedPreferences

You are developing a **Student Management Application**.

The application stores:

- Student ID
- Student Name
- Student Department

Requirements:

1) Write SQLite query to create a table named **students**. [3 Marks]

Constraints:

- id should be **PRIMARY KEY**
- name should **NOT NULL**

2) Write Java code to insert a student record into SQLite database using: [4 Marks]

- SQLiteDatabase
- ContentValues

3) Write Java code to save logged-in student name using SharedPreferences. [3 Marks]

[CLO-1] Question 4 — React Native Components + State Management [10 Marks]
Create a React Native screen named **LoginScreen**.

Requirements:

- Use **useState**
- Create:
 - TextInput for Email
 - TextInput for Password
 - Login Button
- When button is pressed:
 - Display entered email inside Text
- Use:
 - View
 - Text
 - TextInput
 - Button
- Apply basic inline styling

Write complete screen code.



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[10 Marks]

[CLO-2] Question 5: **Firestore Authentication + Intents**

You are developing a **Login System** using Firestore Authentication.

Requirements:

[5 Marks]

a) Write Java code for Firestore Email/Password Login using:

- EditText for email
- EditText for password
- Button for login

[3 Marks]

b) After successful login:

- Open **DashboardActivity**
- Close current activity

using Explicit Intent.

[2 Marks]

c) Display an error message using Toast if login fails.